

The shape of a private pair redistributes its cost: private pairs as a property of the code

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Abstract

The recent lower bounds on the Shannon capacity $\Theta(C_7)$ all rest on gadgets in $C_7^{\boxtimes 5}$: a code, a set of *private pairs* (a vertex q outside the code with a single neighbour r in it), and an auxiliary set that must avoid a forbidden region and is rewarded for avoiding a penalised one. A companion note proves that the Polak–Schrijver code admits exactly eight private pairs. This note explains why the number of pairs is not, by itself, a lever. The two regions a pair creates are the intersection and the union of two closed neighbourhoods, so their sizes always add up to $2 \cdot 3^d$: the shape of a pair moves cost between the forbidden and the penalised region but cannot lower it. We list the five shapes in $C_7^{\boxtimes 5}$ with their costs and the largest independent set we could build carrying each; show that every 367-word code we can reach carries only the two-coordinate shape; that up to automorphism every nine-pair code we can reach has the same sixteen forbidden regions; and that against all of them the best auxiliary set has 366 words, one short, with a status we state precisely: it is 3-opt optimal, and not proved maximum.

1 Context

Since July 2026 every improvement of the lower bound on $\Theta(C_7)$ [6, 5, 4, 10] starts from one five-dimensional gadget [5]: the 367-word Polak–Schrijver code I_0 [9], eight private pairs, two transversals of their endpoints, and an auxiliary set X . The recursions built on it are most sensitive to the number t of private pairs, and a companion note [1] proves that for I_0 this number cannot be raised: $t^*(I_0) = 8$. That leaves the obvious next question — another code with more pairs — and this note answers why that route does not pay as directly as the sensitivity suggests. Together with [1] it accounts for the route “more private pairs” on every code we hold: on I_0 there is no ninth pair, and on the codes that have nine or ten the best auxiliary set falls one word short at nine pairs and two at ten, against forbidden regions that are fixed by the geometry of the pairs rather than by the code. That is a statement about the codes we hold, not about all codes; Section 8 says what would change it.

Only one statement below is proved by hand, and it is elementary. Everything else is a finite computation, exhaustive where we say so and a search where we say so; the status is given with each item, as in [1].

2 Definitions

Vertices of $C_7^{\boxtimes d}$ are the elements of $\mathbb{Z}_7^d$; two distinct vertices are confusable iff their circular distance is at most 1 in every coordinate. $N(S)$ is the *closed* neighbourhood of a set S , as in [5, 10]; for

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a single vertex $|N(v)| = 3^d$. A pair (r, q) with $r \in I$, $q \notin I$ is a *private pair* for an independent set I if $N(q) \cap I = \{r\}$ [5, 10]. A gadget [5] has a code I , private pairs (r_i, q_i) , $1 \leq i \leq t$, with disjoint endpoint sets, two independent complementary transversals P^H, P^V (one endpoint of every pair in each), and an independent X with $X \cap N(P^H) \cap N(P^V) = \emptyset$; its profile counts $o = |X \setminus (N(P^H) \cup N(P^V))|$. We name the two sets in that definition:

$$F = N(P^H) \cap N(P^V) \quad (\text{forbidden}), \quad U = N(P^H) \cup N(P^V) \quad (\text{penalised}).$$

X must avoid F , and every word of X in U is lost from o . The *shape* of a pair is $\delta = q - r \in \{0, \pm 1\}^d \setminus \{0\}$; k is its number of non-zero coordinates.

3 Cost is conserved

Lemma 1. *Let $n \geq 3$ and let P^H, P^V be any two sets of vertices of $C_n^{\boxtimes d}$. Then*

$$|F| + |U| = |N(P^H)| + |N(P^V)| \leq 3^d (|P^H| + |P^V|),$$

with equality iff the closed neighbourhoods of the vertices inside each of P^H and P^V are pairwise disjoint. In particular, for a single pair, $P^H = \{r\}$ and $P^V = \{q\}$,

$$|N(r) \cap N(q)| + |N(r) \cup N(q)| = 2 \cdot 3^d.$$

Proof. $|A \cap B| + |A \cup B| = |A| + |B|$ for any finite sets, with $A = N(P^H)$, $B = N(P^V)$; and $|N(S)| \leq \sum_{s \in S} |N(s)| = 3^d |S|$, since for $n \geq 3$ the three values $a - 1, a, a + 1$ of every coordinate are distinct. $\square$

The lemma is inclusion–exclusion, and it holds for any two vertices, private or not; it is stated because of what it implies for gadgets. A pair contributes $N(r) \cap N(q)$ to F and $N(r) \cup N(q)$ to U , and the two always add up to $2 \cdot 3^d$, which is 486 in $C_7^{\boxtimes 5}$. *The shape of a pair redistributes its cost between the forbidden and the penalised region; it does not lower it.* A shape that makes F small makes U large by the same amount, and U is what o is paid from.

Lemma 2. *For $n \geq 4$ and a pair of shape k , $|N(r) \cap N(q)| = 3^{d-k} 2^k$.*

Proof. Coordinatewise, $\{a - 1, a, a + 1\} \cap \{a + \delta - 1, a + \delta, a + \delta + 1\}$ has 3 elements if $\delta = 0$ and 2 if $\delta = \pm 1$, for $n \geq 4$; closed neighbourhoods in the strong product are products of the coordinate neighbourhoods. $\square$

Both lemmas were also checked by direct count for every shape with $d = 3, 4, 5$ and $n = 3, 4, 5, 7, 9$ ($n = 3$ satisfies Lemma 1 and, as it must, not Lemma 2). For families the inequality is strict in practice: the nine-pair configurations of Section 6 have $|F| + |U|$ between 4050 and 4080 against $2 \cdot 9 \cdot 3^5 = 4374$.

4 The five shapes

The 242 shapes in $C_7^{\boxtimes 5}$ fall into five orbits under the stabiliser of a vertex in $(D_7)^5 \rtimes S_5$, one per k . The last column is a finite computation, exhaustive over its inputs: for every word r of every code we hold and of the swap plateau of Section 6, and every shape, put $q = r + \delta$, delete the other code words confusable with q , and add back an exact maximum independent set of the freed vertices that keep q private. Every set reported is verified, and its pair is re-checked to be private by a second program.

k	shapes	$ N(r) \cap N(q) $ (forbidden)	$ N(r) \cup N(q) $ (penalised)	covered cells in $\text{box}(q)$	largest set found
1	10	162	324	16	366
2	40	108	378	8	367
3	80	72	414	4	366
4	80	48	438	2	366
5	32	32	454	1	366

The cost in F falls with k ; the two-coordinate shape is in the middle, not at the minimum. Every shape is realisable as a private pair, the expensive $k = 1$ included — all seven candidates of the Mathew–Östergård 350-word set [8] have $k = 1$ — but in the sets we can build, shapes other than $k = 2$ occur only one word below 367.

The column of covered cells explains why. A word w occupies the box $w + \{0, 1\}^5$ of 32 cells, and two vertices are non-confusable exactly when their boxes are disjoint; a private q meets the code’s boxes only inside $\text{box}(r)$, in 2^{5-k} cells. A pair of shape $k \geq 3$ therefore needs a box with at most 4 covered cells. In every one of the 367-word codes we can reach, every box outside the code has at least 5.

5 The count is a property of the code

Counting the candidates — the vertices with exactly one neighbour in the code — for every code we hold gives a statistic that, as far as we can find, no paper reports. The counts do not follow size: the linear 343-word code [3] has none, the 350-word set of [8] has seven, the printed 367-word code has eight, and an independent rebuild of it that differs in two words has six. Among the 367-word codes the counts we see run from 5 to 10, every candidate has shape $k = 2$, and for a fixed code the forbidden region varies over the admissible colourings by at most 3.21%.

6 Sixteen forbidden regions

A private pair is also a move: $I \mapsto I - r + q$ keeps size and independence, and it is the exchange used in [10]. Closing every 367-word code we hold under these moves, and under exhaustive exchanges of two and three words, gives a finite plateau that falls into 51 classes under $\text{Aut}(C_7^{\boxtimes 5})$. Each connected component is a product of five independent three-state switches, $3^5 = 243$ codes, and the five switches sit on the five cyclically adjacent pairs of coordinates, permuted by an automorphism of order 5 that cycles the coordinates. (The plateau first appeared with 486 codes, two components; that 486 also equals $2 \cdot 3^5$ is a coincidence: its 3 counts states of a switch and its 2 counts the components our starting codes happened to lie in — a third start adds a third.)

Two classes have nine candidates: the published auxiliary set of [5] read as a code, and one more; one class has ten. An automorphism g moves the forbidden region to $g \cdot F$ and leaves $\alpha(C_7^{\boxtimes 5} \setminus F)$ unchanged, so the variables are the code up to Aut and the colouring. All 208 configurations with $t \geq 9$ give, up to automorphism, exactly 16 forbidden regions at $t = 9$ and 4 at $t = 10$, with

$$1098 \leq |F| \leq 1128 \ (t = 9), \quad 1240 \leq |F| \leq 1268 \ (t = 10),$$

and *each of the sixteen is produced by every nine-pair class*. The forbidden region is a function of the geometry of the nine pairs, which are the same switches in every code, and not of the rest of the code. In the nine-pair regions $972 = 9 \cdot 108$ vertices come from the pairs themselves and 126 to 156 from cross terms between pairs.

7 One word short: what is known, and with what status

Against each of the 20 regions, every image under $(D_7)^5 \rtimes S_5$ of one representative of each of the 51 classes was scored exhaustively: 64,538,880 group elements per pair, 65,829,657,600 images in all. No image avoids any region; the fewest words of an image in F is 1. Every image within three words of admissible was repaired exactly, and the best results are

$$366 \text{ words at } t = 9, \quad 365 \text{ at } t = 10.$$

All 108 repaired 366-word sets are exactly 3-opt optimal inside $G_F = C_7^{\boxtimes 5} \setminus F$: no window of at most three words can be traded for one more, decided by counting blockers rather than by search. Only 2 or 3 allowed vertices are blocked by a single word of such a set.

Status. This is not a proof that 367 words do not fit outside F . The cheap upper bounds cannot see F : the Lovász bound gives 401 whether F is empty or removes 1268 vertices, because it bounds the whole graph, and a box-volume count is weaker still. An honest bound needs ϑ of the induced subgraph on 15,679 to 15,709 vertices, and F breaks the symmetry that would reduce it.

Why a cheaper shape does not obviously help. With $|U|$ taken without overlaps, the fraction of its random share that X may touch at $t = 9$ and 367 words is 0.879, 0.754, 0.688, 0.651, 0.628 for $k = 1, \dots, 5$; the best auxiliary sets on record reach 0.7678. The shapes that would shrink F are the ones that ask most of o (Lemma 1), and at 366 words no shape allows more than 0.68 of the random share at $t = 9, 13$ or 16. These are distances to the best known search, not impossibility results.

8 What remains open

1. *A 367-word code with a deeper hole:* a box covered in at most 4 cells, which a pair of shape $k \geq 3$ needs. None of the codes we can reach has one; finding one requires 367-word codes from outside the known family, which is the open problem around $\alpha(C_7^{\boxtimes 5})$ itself.
2. *An upper bound that sees F ,* of ϑ strength on the induced subgraph, which would turn the one-word gap of Section 7 into a theorem or refute it.

9 Reproducibility

All computations are exact integer computations, published with the scripts that produce them, the predictions registered before each run, the calibrations each estimator had to pass (among them: the new group sweep reproduces the histogram of [1] exactly, the repair reaches 367 at $t = 8$, the window search recovers 367 from deliberately damaged sets, and the automorphism checker rejects a non-automorphism), and a machine reconciliation of every number in this note against the stored results. They were produced through Claude Code under the author's direction.

Data and code availability. Archived with this note on Zenodo [2].

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References

- [1] A. Oktiabrev, The Polak–Schrijver code has exactly eight private pairs, preprint, Zenodo, 2026. doi:10.5281/zenodo.22979509.
- [2] A. Oktiabrev, Sets, verifier and scripts for “The shape of a private pair redistributes its cost” [dataset], Zenodo, 2026. DOI to be assigned on deposit.
- [3] L. Baumert, R. McEliece, E. Rodemich, H. Rumsey, R. Stanley, H. Taylor, A combinatorial packing problem, *Computers in Algebra and Number Theory*, American Mathematical Society, 1971.
- [4] P. Buys, S. Polak, J. Zuiddam, Lean-verified lower bounds for the Shannon capacity of odd cycles, arXiv preprint, 2026. doi:10.48550/arXiv.2607.29681.
- [5] Y. Gao, A recursive construction improving the lower bound on the Shannon capacity of C_7 , arXiv preprint, 2026. doi:10.48550/arXiv.2607.27869.
- [6] N. Itty, C. D. Rosin, C. Carstensen, D. Reichman, Improved lower bounds for the Shannon capacity of odd cycles, arXiv preprint, 2026. doi:10.48550/arXiv.2607.21517.
- [7] L. Lovász, On the Shannon capacity of a graph, *IEEE Transactions on Information Theory* 25 (1979) 1–7.
- [8] K. A. Mathew, P. R. J. Östergård, New lower bounds for the Shannon capacity of odd cycles, *Designs, Codes and Cryptography* 84 (2017) 13–22.
- [9] S. C. Polak, A. Schrijver, New lower bound on the Shannon capacity of C_7 from circular graphs, *Information Processing Letters* 143 (2019) 37–40.
- [10] R. Tandon, Strengthening recursive constructions for zero-error Shannon capacity, arXiv preprint, 2026. doi:10.48550/arXiv.2608.30273.